

1. Administrative & Study Identification

**Full Study Title:** An Open-Label Clinical Study to Evaluate the Effects of Multiple Oral Doses of Rifabutin on the Single-Dose Pharmacokinetics of MK-4646 in Healthy Participants **Short Title/Acronym:** A Study of Rifabutin and MK-4646 in Healthy Participants (MK-4646-004) **Protocol Number:** MK-4646-004 **NCT Number:** NCT07199452 **Posting ID:** 718943246418157113 **Sponsor/Company Name:** Merck (Merck Sharp & Dohme LLC) **Investigational Product:** MK-4646 (Investigational HIV drug) and Rifabutin **Phase of Development:** Phase I **Trial Status:** Recruiting **Medical Monitor:** Merck Clinical Director

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2. Study Synopsis & Rationale

**2.1 Study Objective Primary Objective:** To evaluate the effects of multiple oral doses of Rifabutin on the single-dose pharmacokinetics of MK-4646 in healthy participants. **Secondary Objectives:** To evaluate the safety and tolerability of MK-4646 when administered as a single dose alone and in combination with multiple doses of Rifabutin.

**2.2 Background & Rationale** To address potential drug-drug interactions between MK-4646, a new investigational drug under development for treating human immunodeficiency virus (HIV), and Rifabutin, an established medication used to treat tuberculosis (TB). Understanding how MK-4646 interacts with Rifabutin is essential to determine if this combination could be safely co-administered in patients co-infected with HIV and TB.

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3. Study Design & Methodology

**3.1 Design Framework Study Type:** Interventional **Control Type:** Self-controlled (crossover-style sequential design) **Blinding:** Open-label **Randomization:** N/A (Single-sequence)

**3.2 Planned Enrollment & Duration Target \$N\$:** 18 **Number of Sites:** 1 (Phase 1 clinical pharmacology unit) **Estimated Study Start:** 28-Oct-2025 **Estimated Primary Completion:** 27-Dec-2025

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4. Subject Selection (Eligibility Criteria)

**4.1 Inclusion Criteria** 1. Is in good health. 2. Between the ages of 18 and 55 years. 3. Has a body-mass index (BMI) between 18 and 32 \$kg/m^2\$.

**4.2 Exclusion Criteria** 1. Has a history of clinically significant endocrine, gastrointestinal, cardiovascular, hematological, hepatic, immunological, renal, respiratory, genitourinary, or major neurological (including stroke and chronic seizures) abnormalities or diseases. 2. Has a history of cancer (malignancy).

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**5. Intervention & Treatment Plan**

**5.1 Investigational Regimen Dosage/Frequency:** \* **Period 1:** A single oral dose of MK-4646 administered on Day 1. \* **Period 2:** Rifabutin (Trade names: Mycobutin, Talicia | ATC Code: J04AB04) administered orally once daily from Day 1 through Day 16. A single oral dose of MK-4646 administered on Day 14. **Route of Administration:** Oral **Duration of Treatment:** Approximately 3 weeks of active treatment periods.

**5.2 Concomitant & Prohibited Therapy Permitted:** N/A (Standard Phase 1 restrictions apply for healthy volunteers). **Prohibited:** Other medications or supplements that may interfere with CYP enzymes or interact with Rifabutin/MK-4646 pharmacokinetics.

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**6. Study Timeline & Visit Schedule**

|          |        |                           |                |                                                                                     |      |
|----------|--------|---------------------------|----------------|-------------------------------------------------------------------------------------|------|
| Visit ID | Window | Key Activities/Procedures |                | ----                                                                                | ---- |
| ----     |        | <b>Screening</b>          | Days -28 to -1 | Informed Consent, Medical History, BMI verification, Eligibility checks             |      |
|          |        | <b>Period 1</b>           | Day 1          | Single oral dose of MK-4646, PK blood sampling                                      |      |
|          |        | <b>Period 2</b>           | Days 1-16      | Daily Rifabutin administration; MK-4646 administration on Day 14, PK blood sampling |      |
|          |        | <b>End of Study</b>       |                | Follow-up                                                                           |      |
|          |        |                           |                | Final Safety Labs, Exit Evaluation, AE monitoring                                   |      |

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**7. Outcome Measures (Endpoints)**

**7.1 Primary Endpoint Definition:** Area Under the Concentration-time Curve from Time 0 Extrapolated to Infinity (\$AUC0-inf\$) of MK-4646 in Plasma. **Measurement Method:** Blood samples collected to determine the \$AUC0-inf\$ of MK-4646 in plasma.

**7.2 Secondary Endpoints** 1. Number of participants who experience an Adverse Event (AE). 2. Number of participants who discontinue study treatment due to an AE.

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**8. Safety & Adverse Event Reporting**

**Adverse Event (AE) Monitoring:** Standard Phase 1 clinical evaluation, vital signs, and laboratory assessments during inpatient/outpatient study visits. **Serious Adverse Events (SAEs):**

Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by the sponsor (Merck).

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## 9. Statistical Analysis Plan (SAP) Summary

**Analysis Populations:** Pharmacokinetic (PK) Analysis Set for primary endpoints; Safety Analysis Set for all AE/SAE evaluations. **Sample Size Justification:** Sized to provide adequate precision for estimating the ratio of geometric means for MK-4646 PK parameters (with and without Rifabutin). **Handling of Missing Data:** Standard Phase 1 PK non-compartmental analysis rules (e.g., missing concentrations excluded or imputed as zero if pre-dose).

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## 10. Quality Assurance & Regulatory Compliance

**GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Continuous monitoring during confinement and study visits at the Phase 1 facility. **Audit Trail:** Electronic data capture (EDC) system with strict audit trails for PK sample tracking and lab results.

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## 11. Study Identifiers & Governance

**Primary ID:** MK-4646-004 **Registry Number:** NCT07199452 **Sponsor/Lead Organization:** Merck (Merck Sharp & Dohme LLC) **Study Title:** A Study of Rifabutin and MK-4646 in Healthy Participants **Official Regulatory Title:** An Open-Label Clinical Study to Evaluate the Effects of Multiple Oral Doses of Rifabutin on the Single-Dose Pharmacokinetics of MK-4646 in Healthy Participants

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## 12. Clinical Framework (Study Specific)

**Primary Condition:** Healthy Volunteers (Disease Area: Healthy) **Diagnosis Threshold:** Good health verified by standard Phase 1 clinical and lab assessments **Medical Methodology:** Pharmacokinetic (PK) drug-drug interaction (DDI) assessment **Optimization Target:** Changes in MK-4646 plasma concentrations (\$AUC<sub>0-inf</sub>\$)

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## 13. Study Procedures & Methodology

**Management Pathway:** Standard Phase 1 clinical pharmacology unit protocols **Education Protocol (HCP):** Site initiation and rigorous protocol training for PK sampling **Education Protocol (Patient):** Informed consent, Phase 1 unit rules, and dietary/activity

restrictions **Quality Audit Mechanism:** Source data verification and strict PK sample temperature/time tracking

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## 14. Observation & Follow-Up Schedule

**Total Duration:** Approximately 3-4 weeks (Screening + Dosing periods) **Onsite Visit Cadence:** Daily during dosing periods (in-patient confinement or scheduled outpatient visits) **Remote Monitoring Frequency:** N/A (Direct observation setting) **Checklist Review Interval:** Daily check-ins for AE monitoring and dosing compliance

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## 15. Sign-off & Approval

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|--------------------------|--------|-----------|---------------|-----|-----|------|-----|
| Role                     | Name   | Signature | Date          | :-- | :-- | :--  | :-- |
| Principal Investigator   | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |
| Clinical Operations Lead | [Name] | _____     | [DD-MMM-YYYY] |     |     | Lead |     |
| Statistician             | [Name] | _____     | [DD-MMM-YYYY] |     |     |      |     |